

Training-Free One-to-Many Video Temporal Grounding via Sub-Linear Duration-Adaptive Windowing

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Abstract. Multimodal Large Language Models (MLLMs) are typically evaluated on single-event retrieval. When evaluated on One-to-Many Video Temporal Grounding (OMTG), where a natural language query corresponds to multiple disjoint intervals in long videos, standard models struggle to detect multiple occurrences. In this paper, we propose Adaptive Scout-Guided Dense Evidence (Adaptive SGDE), a training-free framework that improves multi-event temporal localization without parameter fine-tuning. We show that baseline performance drops stem largely from prompt formatting issues and uninformative background frames in long videos. By combining structured prompt formatting with a lightweight 1.04B scout and duration-adaptive window planning, our approach removes unnecessary background frames before dense inference. On the OMTG benchmark, Adaptive SGDE achieves 33.12% Count Accuracy and 19.86% Effective Temporal F1, while reducing primary 8B model frame evaluations by 60.3%.

Keywords: Video Temporal Grounding · Multimodal LLMs · One-to-Many Grounding · Duration-Adaptive Windowing · Training-Free Inference.

1 Introduction

Video Temporal Grounding (VTG) aims to identify start and end timestamps in untrimmed video corresponding to a natural language query [1, 2]. While traditional benchmarks focus on the single-instance setting where each query matches exactly one moment, real-world videos frequently contain recurring actions. In domains such as surveillance, sports analytics, and video search, an action described by a query often repeats multiple times across non-contiguous intervals.

To evaluate this capability, recent work introduced One-to-Many Video Temporal Grounding (OMTG) [10]. Unlike traditional single-span retrieval, OMTG requires both event counting and boundary localization for each event occurrence. Standard Multimodal Large Language Models (MLLMs), including Gemini-1.5-Pro, Video-LLaVA [7], and TimeLens-8B [4], struggle on OMTG, obtaining 0.00% Count Accuracy and 0.00% Effective Temporal F1 under standard prompts in official evaluations [10].

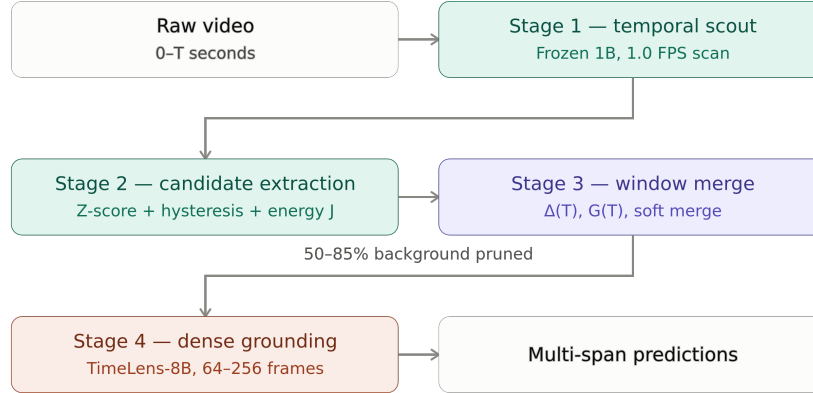


Fig. 1. Overview of Adaptive Scout-Guided Dense Evidence (Adaptive SGDE). Long untrimmed videos are first scanned by a lightweight 1.04B scout at 1.0 FPS. Local Z-score normalization and thresholding extract candidate action proposals, which are grouped by duration-adaptive windowing and continuity merging. The focused candidate window is decoded with dense frames by a frozen 8B Vision-Language model using schema-delimited JSON syntax, removing background frames while preserving the number of events.

In this paper, our empirical analysis highlights two major contributing factors behind the low multi-event grounding performance of MLLMs:

1. **Prompt Format Misalignment:** Default benchmark prompts embed output instructions within unstructured dialogue. This causes instruction-tuned models to output conversational text or a single timestamp instead of structured multi-span lists.
2. **Background Frames in Long Videos:** In long untrimmed videos spanning several minutes, uniform sampling forces the Vision Transformer to process thousands of irrelevant background frames. These background frames can introduce irrelevant visual evidence and reduce grounding accuracy.

To address both issues without parameter fine-tuning, we propose Adaptive Scout-Guided Dense Evidence (Adaptive SGDE), a training-free framework for multi-event temporal grounding. As shown in Fig. 1, Adaptive SGDE uses a four-stage coarse-to-fine pipeline:

- **Stage 1: Coarse Multimodal Scouting:** A frozen 1.04B embedding scout scans the untrimmed video at 1.0 FPS, producing a cross-modal similarity timeline normalized by video-level Z-scores because similarity scores vary across videos.
- **Stage 2: Proposal Extraction:** Dual-threshold search and energy scoring extract candidate action intervals.

- **Stage 3: Duration-Adaptive Windowing:** Sub-linear context margins and clustering gaps group candidates across diverse video lengths, unified by a continuity merge rule to maintain cross-event context.
- **Stage 4: Focused Dense Grounding:** The primary grounder (TimeLens-8B [4]) processes only the candidate window under a structured JSON prompt, focusing frame allocation on relevant action regions.

Evaluations on the OMTG benchmark demonstrate that Adaptive SGDE improves training-free grounding, achieving 33.12% Count Accuracy and 19.86% Effective Temporal F1. Furthermore, by removing irrelevant background intervals, Adaptive SGDE reduces 8B Vision Transformer frame evaluations by 60.3%.

2 Related Work

2.1 Video Temporal Grounding: Single-Event to One-to-Many

Video Temporal Grounding (VTG) aims to localize temporal boundaries in an untrimmed video corresponding to a natural language query [1, 2]. Early approaches explored cross-modal proposal ranking [1] and anchor regression [6]. The introduction of the QVHighlights benchmark [5] expanded the field toward joint moment retrieval and highlight detection. However, standard datasets such as Charades-STA [1] and ActivityNet-Captions [2] enforce a single-instance setting. To evaluate recurring actions, Xu *et al.* [10] introduced One-to-Many Temporal Grounding (OMTG), revealing that standard video grounders and multimodal LLMs struggle on multi-instance queries. In this work, we present a training-free framework designed specifically for multi-event grounding.

2.2 Multimodal Large Language Models for Video Understanding

Recent Multimodal Large Language Models (MLLMs), such as Video-LLaVA [7], VTimeLLM [3], Qwen2-VL [9], and TimeLens-8B [4], demonstrate strong zero-shot temporal comprehension. TimeLens-8B incorporates dedicated coordinate tokens to predict continuous timestamps in text. Nonetheless, when processing long untrimmed videos, standard MLLMs sample visual tokens uniformly across the entire video. In long videos with extended background segments, uniform sampling dilutes visual attention and causes false positive predictions in empty intervals. Our work shows that restructuring prompt syntax recovers latent multi-span ability, while candidate window extraction removes irrelevant temporal regions.

2.3 Training-Free and Coarse-to-Fine Video Grounding

Processing long videos efficiently has motivated hierarchical coarse-to-fine strategies [6, 8]. In contrast to approaches requiring gradient fine-tuning or model changes, training-free methods route computation dynamically at inference time. Adaptive SGDE uses a lightweight 1.04B embedding scout operating at 1.0 FPS

to scan the timeline, followed by duration-adaptive windowing and continuity merging. This decouples coarse temporal search from fine boundary decoding, reducing 8B Vision Transformer frame evaluations by 60.3% while improving localization and count accuracy.

3 Methodology: Adaptive SGDE

3.1 Problem Formulation and Overview

Given an untrimmed video $\mathcal{V} = \{f_t\}_{t=1}^T$ of duration T seconds and a natural language query $\mathcal{Q}$, One-to-Many Video Temporal Grounding (OMTG) [10] requires localizing all $K \geq 1$ disjoint action intervals:

$$\mathcal{Y}^* = \{[s_k^*, e_k^*]\}_{k=1}^K, \quad \text{subject to } 0 \leq s_1^* < e_1^* < s_2^* < \dots < e_K^* \leq T, \quad (1)$$

where each temporal segment $[s_k^*, e_k^*]$ semantically satisfies $\mathcal{Q}$. OMTG requires predicting the total event count K as well as start and end timestamps $[s_k, e_k]$ for each retrieved moment. Standard Multimodal Large Language Models (MLLMs) perform poorly on OMTG due to background frames in long videos and prompt format misalignment.

To address these challenges without fine-tuning, we propose Adaptive Scout-Guided Dense Evidence (Adaptive SGDE). As illustrated in Fig. 1, Adaptive SGDE uses a four-stage coarse-to-fine pipeline: (1) 1.0 FPS timeline scouting (§3.2), (2) proposal extraction and scoring (§3.3), (3) duration-adaptive window planning (§3.4), and (4) budgeted dense decoding with structured multi-span output (§3.5).

3.2 Stage 1: Coarse-Grained Temporal Timeline Scouting

Lightweight Multimodal Scouting. Feeding unpruned raw video into an 8B Vision Transformer requires evaluating large numbers of irrelevant background frames. We instead deploy a lightweight, frozen 1.04B scout $\mathcal{M}_{\text{scout}}$ (Llama-Nemotron-Embed-VL-1B) at a uniform coarse rate of 1.0 FPS. For each timestamp $t \in \{1, \dots, T\}$, we extract frame embedding $\mathbf{v}(t) = \mathcal{M}_{\text{scout}}(f_t)$ and query embedding $\mathbf{q} = \mathcal{M}_{\text{scout}}(\mathcal{Q})$, computing continuous cross-modal cosine similarity:

$$S(t) = \frac{\langle \mathbf{v}(t), \mathbf{q} \rangle}{\|\mathbf{v}(t)\|_2 \|\mathbf{q}\|_2}, \quad t \in \{1, \dots, T\}. \quad (2)$$

Per-Video Similarity Normalization. Because similarity scores vary across videos, we normalize the similarity scores separately for each video using Z-score normalization:

$$Z(t) = \frac{S(t) - \mu_S}{\sigma_S + \epsilon}, \quad \text{where } \mu_S = \frac{1}{T} \sum_{t=1}^T S(t), \quad \sigma_S = \sqrt{\frac{1}{T} \sum_{t=1}^T (S(t) - \mu_S)^2}, \quad (3)$$

with numerical stability constant $\epsilon = 10^{-6}$. The resulting timeline $Z(t)$ standardizes scores across videos, enabling consistent peak and energy thresholding.

3.3 Stage 2: Candidate Proposal Extraction and Scoring

Dual-Threshold Search. Single-threshold filtering faces a trade-off: high thresholds fragment extended actions into disjoint segments, while low thresholds introduce spurious background detections. We use a dual-threshold search with trigger threshold $\tau_{\text{high}} = 0.80$ and expansion threshold $\tau_{\text{low}} = 0.25$. An action candidate $c = [a, b]$ is initiated only if it contains a peak $Z(t_p) \geq \tau_{\text{high}}$, and is grown bidirectionally until the score falls below τ_{low} .

Proposal Scoring. To score candidate proposals without length bias, we compute the average excess similarity energy $\bar{J}(a, b)$:

$$\bar{J}(a, b) = \frac{1}{b - a + 1} \sum_{t=a}^b \max(0, Z(t) - \tau_{\text{base}}), \quad (4)$$

where baseline offset $\tau_{\text{base}} = 0.30$. Each candidate $c = [a, b]$ is assigned a composite score:

$$\text{Score}(c) = \max_{t \in [a, b]} Z(t) + \alpha_{\text{energy}} \cdot \bar{J}(a, b), \quad (5)$$

with $\alpha_{\text{energy}} = 0.50$. Applying 1D Non-Maximum Suppression (NMS) with threshold $\theta_{\text{nms}} = 0.30$ removes redundant overlapping proposals.

3.4 Stage 3: Duration-Adaptive Window Planning

Adaptive Margin and Clustering Gap. Fixed temporal windows do not scale well across diverse video durations, over-expanding short clips and truncating long videos. We introduce duration-adaptive geometry governed by sub-linear functions:

$$\Delta(T) = \text{clamp}(3.5 \ln T, 8.0, 22.0) \text{ s}, \quad G(T) = \max(15.0, 3.5\sqrt{T}) \text{ s}, \quad (6)$$

where $\Delta(T)$ denotes the context margin and $G(T)$ is the clustering gap. The margin $\Delta(T)$ provides an 8.0s context buffer for 30s clips while expanding to 22.0s for long videos. Adjacent proposals separated by distance $\leq G(T)$ are clustered into candidate windows.

Window Merging. Action comprehension degrades when instances from the same video are split into disjoint prompts. For candidate envelope $W_{\text{start}} = \max(0, \min_i(a_i - \Delta(T)))$ and $W_{\text{end}} = \min(T, \max_i(b_i + \Delta(T)))$ clamped to video boundaries $[0, T]$, we merge intervals into a single continuous window if the envelope span ratio satisfies:

$$\frac{W_{\text{end}} - W_{\text{start}}}{T} \geq \rho = 0.70, \quad \text{or adjacent gap} \leq G(T). \quad (7)$$

This merge rule preserves contextual continuity across proximate actions while removing uninformative background frames outside the candidate envelope. The parameter ρ controls the trade-off between prompt continuity and background pruning, which we analyze in §4.3.

3.5 Stage 4: Dense Grounding and Structured Multi-Span Decoding

Frame Allocation. Within the candidate window $[W_{\text{start}}, W_{\text{end}}]$, we allocate a fixed frame budget $B \in \{64, 128, 256\}$. Sampling B frames inside the candidate window yields an effective local frame rate of $\text{FPS}_{\text{local}} = B/(W_{\text{end}} - W_{\text{start}})$, which concentrates visual resolution on the candidate region compared to uniform whole-video sampling.

Schema-Delimited Multi-Span Prompting. Unstructured prompts often cause generation issues. We format the inference query into a structured JSON prompt:

Given the query: "{query}", return ALL time spans (in seconds) where the query is relevant. Output format MUST be a JSON array of [start, end] pairs: [[s1, e1], [s2, e2], ...].

The primary grounder $\mathcal{M}_{\text{ground}}$ (TimeLens-8B [4]) decodes exact boundary timestamps $\mathcal{Y} = \{[s_k, e_k]\}_{k=1}^M$ over the densely sampled candidate window. When candidate proposals form widely separated clusters ($\frac{W_{\text{end}} - W_{\text{start}}}{T} < \rho$), disjoint windows are prompted independently with frame budget B , and their resulting predicted JSON timestamp lists are concatenated and chronologically sorted.

4 Experiments

4.1 Experimental Setup

Benchmark Dataset. We evaluate on the complete **OMTG Bench** [10], the established benchmark for One-to-Many Video Temporal Grounding. The benchmark consists of 320 manually verified query-video pairs spanning 287 untrimmed videos from Charades, ActivityNet, QVHighlights, VTimeLLM, and Moment10M. Durations range from 12.0s to 506.0s (mean: 161.4s), with ground-truth instance counts ranging from 2 to 20 per query (mean: 3.67 instances).

Evaluation Metrics. Following the official OMTG evaluation protocol [10], we report:

1. **Count Accuracy (C-Acc, %)** $\uparrow$: Percentage of samples where the predicted segment count exactly matches ground-truth cardinality ($M = K$).
2. **Temporal F1 (tF1@ ξ , %)** $\uparrow$: Harmonic mean of precision and recall under Hungarian bipartite matching at IoU thresholds $\xi \in \{0.3, 0.5, 0.7\}$, with unmatched predictions penalized as false alarms.
3. **Temporal IoU (tIoU, %)** $\uparrow$: Mean temporal intersection-over-union across matched ground-truth and predicted intervals.
4. **Effective Temporal F1 (EtF1, %)** $\uparrow$: The primary benchmark metric, which conditions temporal F1 on exact cardinality match ($\text{EtF1} = \text{tF1} \times \mathbf{1}(M = K)$).
5. **Cardinality Error** $\downarrow$: Mean absolute error between predicted and ground-truth event counts ($|M - K|$).
6. **Processed Frames and Compute Saved (%)** $\uparrow$: Aggregate video frames processed by the 8B Vision Transformer across all 320 evaluations ($N_{\text{frames}} = \sum_i F_i$), and relative frame reduction Saved = $(1 - N_{\text{frames}}^{\text{method}} / N_{\text{frames}}^{\text{native}}) \times 100\%$.

Table 1. Master performance and efficiency comparison on OMTG Bench ($N = 320$). All models operate zero-shot without fine-tuning. Frames denotes aggregate 8B ViT frames evaluated across the benchmark. Rows with 128 frames share identical structured prompt schemas for strictly controlled comparison. Best results are in **bold**; second-best are underlined.

Configuration	Frames	Saved $\uparrow$	C-Acc $\uparrow$	tF1@.5	tIoU $\uparrow$	EtF1 $\uparrow$	Err $\downarrow$
TimeLens-8B (Paper Prompt) [10]	103.3k	Baseline	0.00%	32.76%	32.38%	0.00%	2.15
TimeLens-8B (JSON Prompt)	103.3k	Baseline	19.69%	47.98%	46.57%	14.23%	2.07
+ Naive Uniform Downsample (128f)	41.0k	-60.3%	16.88%	45.30%	44.23%	12.29%	1.96
+ Adaptive SGDE-64 (64f)	20.5k	-80.2%	27.19%	37.41%	41.22%	14.44%	2.09
+ Adaptive SGDE-128 (128f) [Proposed]	41.0k	-60.3%	33.12%	41.82%	44.76%	<u>19.14%</u>	1.75
+ Adaptive SGDE-256 (256f)	81.9k	-20.7%	31.88%	41.82%	<u>45.42%</u>	19.86%	<u>1.81</u>
Oracle Ground-Truth Window (128f)	41.0k	-60.3%	33.44%	<u>47.59%</u>	50.54%	20.99%	2.08

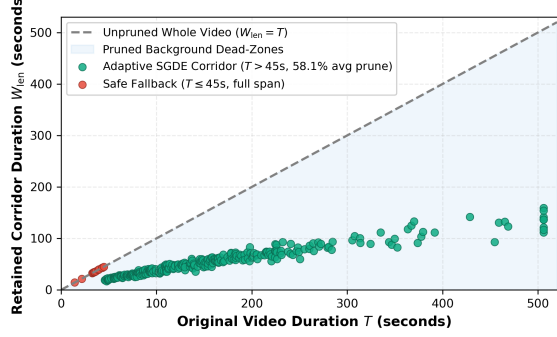


Fig. 2. Original video duration vs. retained candidate window duration on OMTG Bench ($N = 320$). The diagonal dashed line ($W_{len} = T$) denotes the unpruned baseline. For short videos ($T \leq 45s$), Adaptive SGDE retains the full video span (0% pruned, red points). For longer videos ($T > 45s$), candidate window extraction removes an average of 58.1% of background frames (green points, shaded area).

Implementation Details. Our primary grounder is frozen TimeLens-8B [4] (Qwen3-VL-8B-Instruct backbone). The temporal scout is frozen Llama-Nemotron-Embed-VL-1B (1.04B parameters) sampled at a coarse 1.0 FPS. Hyperparameters are set prior to benchmark evaluation and kept fixed across all main experiments: detection thresholds $\tau_{high} = 0.80, \tau_{low} = 0.25$ (§3.3), energy baseline offset $\tau_{base} = 0.30$ (Eq. 4), adaptive window coefficients $\alpha_{margin} = 3.5s, \beta_{gap} = 3.5s$ (Eq. 6), and continuity merge threshold $\rho = 0.70$ (Eq. 7). We report a post-hoc sensitivity analysis of ρ in §4.3.

4.2 Grounding Performance and Computational Efficiency

Table 1 and Fig. 2 present grounding accuracy and compute efficiency on OMTG Bench. Four primary insights emerge:

1. Prompt Syntax vs. Grounding Capacity. Under the unstructured prompt used in the official OMTG study [10], TimeLens-8B outputs conversational fragments or single moments, resulting in 0.00% Count Accuracy and 0.00% Effective Temporal F1. Enforcing schema-delimited JSON syntax restores multi-span output, reaching 19.69% Count Accuracy and 14.23% Effective Temporal F1. This indicates that baseline failure in prior benchmarks was partly caused by prompt formatting rather than model capacity.

2. Video Length Scaling and Background Removal. As shown in Fig. 2, benchmark videos range from 12.0s to 506.0s. On long unpruned videos, background frames can introduce irrelevant visual evidence and reduce grounding accuracy. Adaptive SGDE preserves short videos while removing an average of **58.1% of background frames** on longer videos. Removing these background segments enables Adaptive SGDE-128 to increase Count Accuracy to **33.12%** and reduce Cardinality Error from 2.07 to **1.75**.

3. Comparison with Oracle Ground-Truth Window Reference. We use the ground-truth window as an oracle reference to measure the performance of the grounder when the relevant temporal region is known. As reported in Table 1, sampling 128 frames within this oracle window achieves 33.44% Count Accuracy and 20.99% Effective Temporal F1. Without access to ground-truth temporal boundaries, training-free Adaptive SGDE-128 achieves **33.12% vs. 33.44% Count Accuracy** and **19.14% vs. 20.99% Effective Temporal F1**, demonstrating that the lightweight scout effectively selects relevant event regions.

4. Multi-Event Localization and Compute Savings. The JSON prompt baseline predicts fewer event instances, resulting in 19.69% Count Accuracy and a Cardinality Error of 2.07. By focusing frame allocation on candidate action intervals, Adaptive SGDE substantially improves multi-event detection, achieving **33.12% Count Accuracy**, reducing Cardinality Error to **1.75**, and reaching **19.86% Effective Temporal F1** with 256 frames. Concurrently, focusing 8B model decoding on candidate windows reduces frame evaluations by **60.3%**, offloading coarse temporal search to the lightweight 1.04B scout.

4.3 Algorithmic Analysis: Window Planning and Windowing Topologies

1. Semantic Selection vs. Random Windowing. As shown in Table 2, evaluating a random temporal window matching the average candidate duration ratio of 40% drops Effective Temporal F1 to **3.61%** and tIoU to **24.18%**. Random temporal windowing performs poorly, showing that selecting a window without semantic selection is not sufficient.

2. Single-Window vs. Multi-Window Topology. Moving from raw video to a single continuous window (strategy b) improves Count Accuracy from 16.88% to **28.44%** and Effective Temporal F1 from 12.29% to **16.86%**. However, a single

Table 2. Ablation of temporal window planning and windowing strategies on OMTG Bench ($N = 320$, 128-frame budget). Isolating semantic scouting, single-window bounding, and soft continuity merging.

Window Strategy	Planning Mode	C-Acc $\uparrow$	tF1@.5	tIoU $\uparrow$	EtF1 $\uparrow$	Err $\downarrow$
(a) Random Temporal Crop	Random window (40% of video duration)	18.44%	24.29%	24.18%	3.61%	2.02
(b) Single-Window Baseline	Single bounding window	28.44%	38.26%	36.81%	16.86%	1.94
(c) Multi-Window (No Merge)	Split on gaps ($\rho = 1.0$)	28.44%	38.49%	37.12%	16.81%	1.94
(d) Adaptive SGDE-128 (Full)	Adaptive window + merge ($\rho = 0.70$)	33.12%	41.82%	44.76%	19.14%	1.75

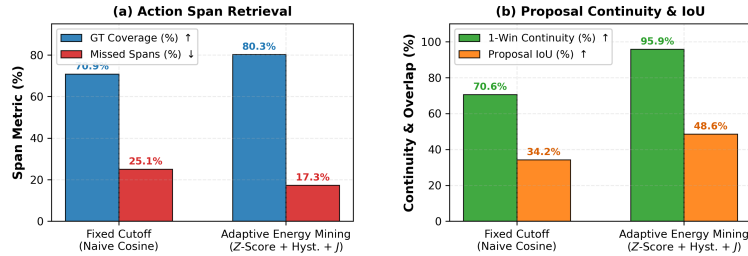


Fig. 3. Evaluation of timeline scoring and candidate extraction methods on OMTG Bench. (a) Action span retrieval: Z-score normalization and energy scoring increase Ground-Truth coverage to 80.3% and cut missed spans by 31.1% relative. (b) Proposal continuity and overlap: achieving 95.9% 1-window continuity and 48.6% proposal IoU.

window can include large background regions between distant events, limiting tIoU to 36.81%. While splitting intervals without continuity merging (strategy c) can fragment adjacent actions, our proposed Adaptive SGDE with 70% continuity merging (strategy d) groups proximate clusters while isolating distant intervals, improving tIoU to **44.76%**, Count Accuracy to **33.12%**, and Effective Temporal F1 to **19.14%**.

3. Timeline Scoring Function Evaluation. As shown in Fig. 3, raw cosine similarity with a fixed cutoff is sensitive to baseline shifts across videos, resulting in 25.1% missed action spans. Video-adaptive Z-score normalization with dual-threshold search and energy scoring $\bar{J}(a, b)$ improves ground-truth coverage from 70.9% to **80.3%**, improves proposal IoU from 34.2% to **48.6%**, and reduces missed action spans by 31.1% relative.

4. Sensitivity Analysis of Merge Threshold (ρ). We set $\rho = 0.70$ prior to benchmark evaluation and examine its sensitivity post-hoc in Fig. 4. Low thresholds ($\rho \leq 0.40$) merge almost all proposals into a single window, maintaining continuous context but pruning fewer background frames (33.3%). Conversely, high thresholds ($\rho \geq 0.90$) or gap-only splitting divide multiple videos into disjoint prompts, which degrades relational context and lowers Effective Temporal F1 to 16.81%. Setting $\rho = 0.70$ provides an effective empirical trade-off: maintain-

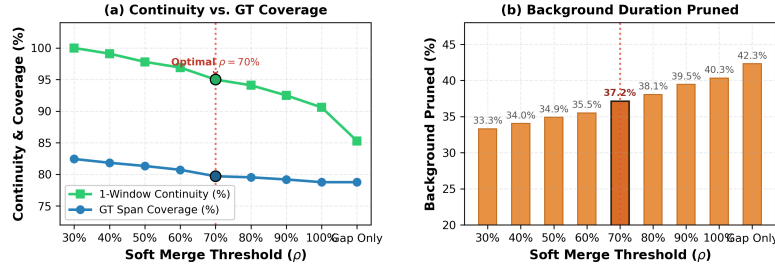


Fig. 4. Merge threshold (ρ) sweep on OMTG Bench. (a) Continuity vs. GT coverage: setting $\rho = 70\%$ retains 95.0% of videos within a single continuous prompt while covering 79.71% of GT action spans. (b) Background duration pruned: achieving 37.17% background pruning across the benchmark.

ing a single continuous window for 95.0% of videos while capturing 79.71% of ground-truth action spans and removing 37.17% of background frames overall.

5 Conclusion

In this paper, we addressed the multi-event temporal grounding problem in Multimodal Large Language Models. We showed that baseline performance drops stem from prompt format misalignment and uninformative background frames in long videos. To address both challenges without fine-tuning, we presented Adaptive Scout-Guided Dense Evidence (Adaptive SGDE), a four-stage training-free framework combining temporal scouting at 1.0 FPS, per-video similarity normalization, energy scoring, duration-adaptive windowing, and structured JSON decoding. On the complete OMTG benchmark, Adaptive SGDE achieves 33.12% Count Accuracy and 19.86% Effective Temporal F1, while reducing 8B model frame evaluations by 60.3%.

Limitations and Future Work. At 1.0 FPS, our proposal extraction covers 80.3% of ground-truth action spans, which limits the current system on very brief actions. Future work will investigate adaptive scout sampling rates and in-ViT motion-residual token pruning for scaling to hour-long video comprehension.

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